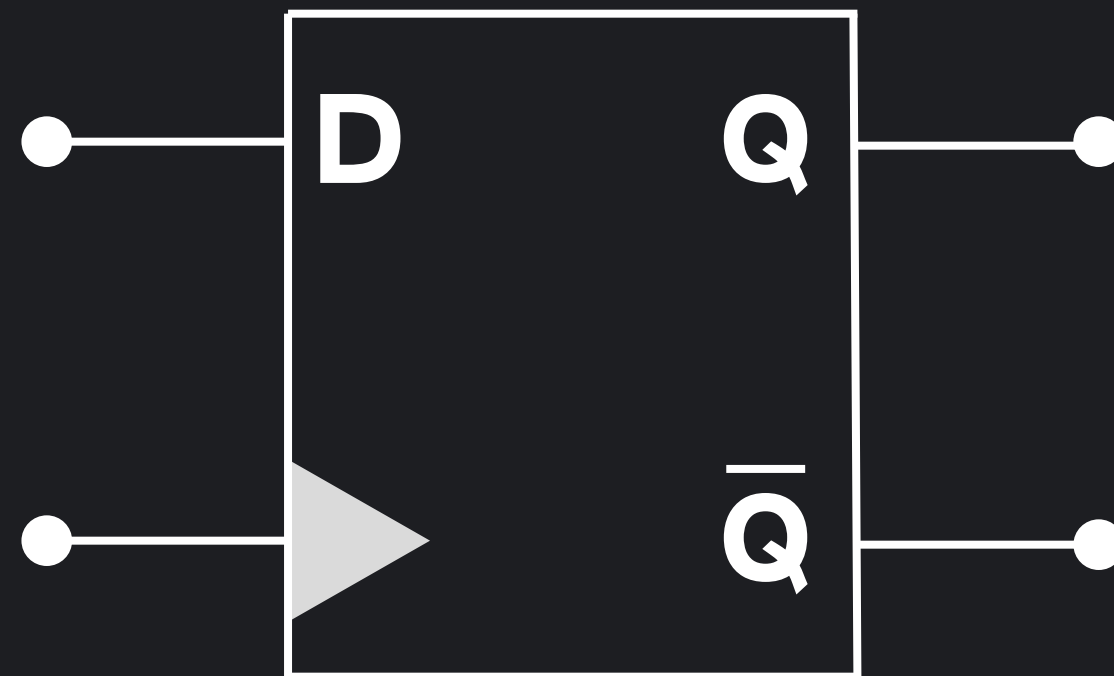


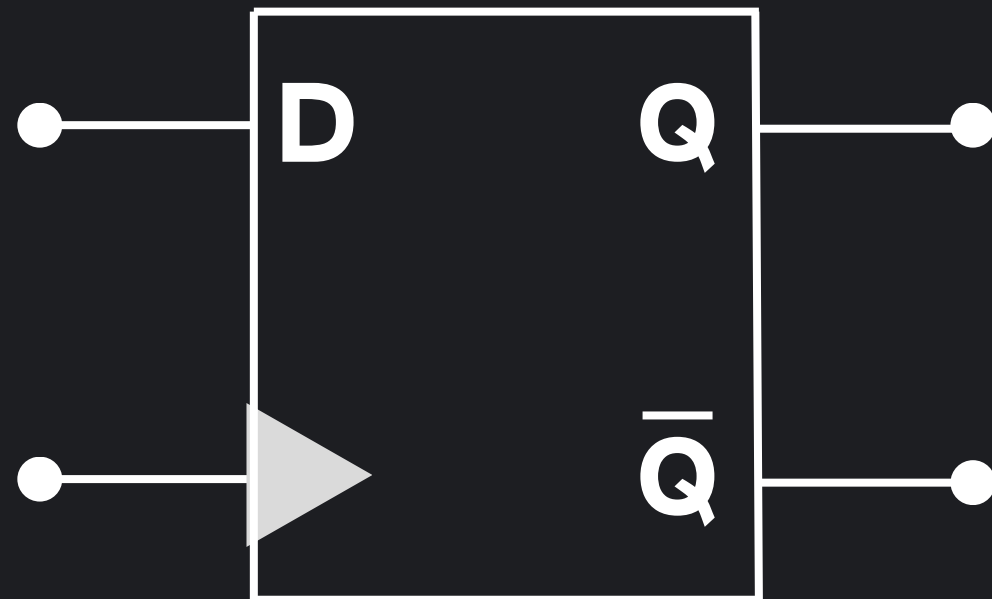
D-FLIP FLOP



What is a D-FLIP FLOP?

A D Flip-Flop is a basic sequential logic gate that produces an output that matches its input at the time of a clock pulse. The output is in a HIGH (1) state when the Data input (D) is HIGH (1), while it is in a LOW (0) state when the input is LOW (0).

Truth Table



D	CK	Q	made of operation
x	↓	Q_0	NO Change
0	↑	0	Resets
1	↑	1	Sets

Function of the D-FLIP FLOP

A D Flip-Flop performs a logical data transfer and delay operation on its inputs. The main function of a D Flip-Flop is to store the value of the data input at a specific moment and produce an output signal according to this rule:

Output = 1 (Set) when the data input $D = 1$ at the clock pulse.

Output = 0 (Reset) when the data input $D = 0$ at the clock pulse.

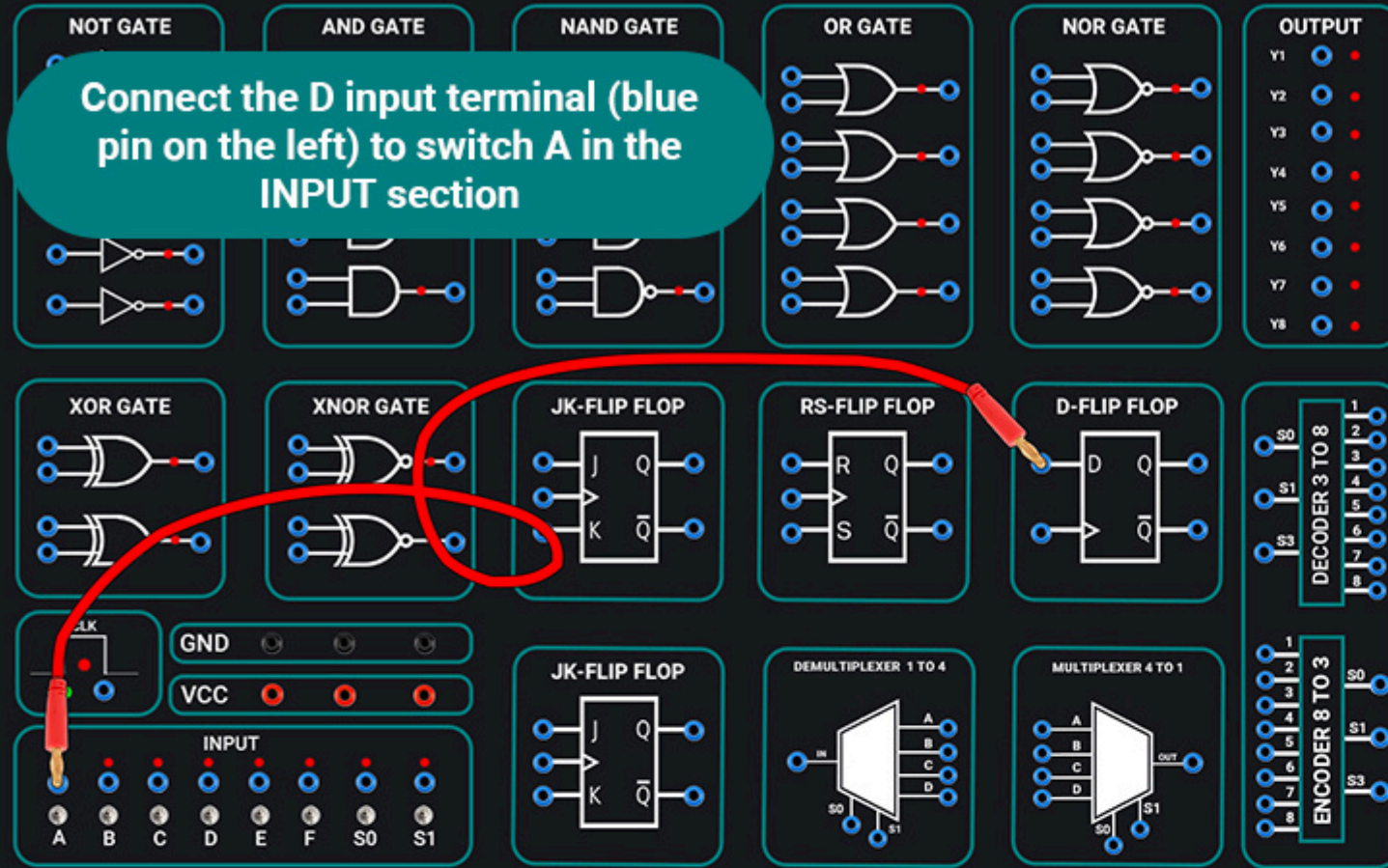
Output = No Change when there is no clock pulse (regardless of D value).

-

1

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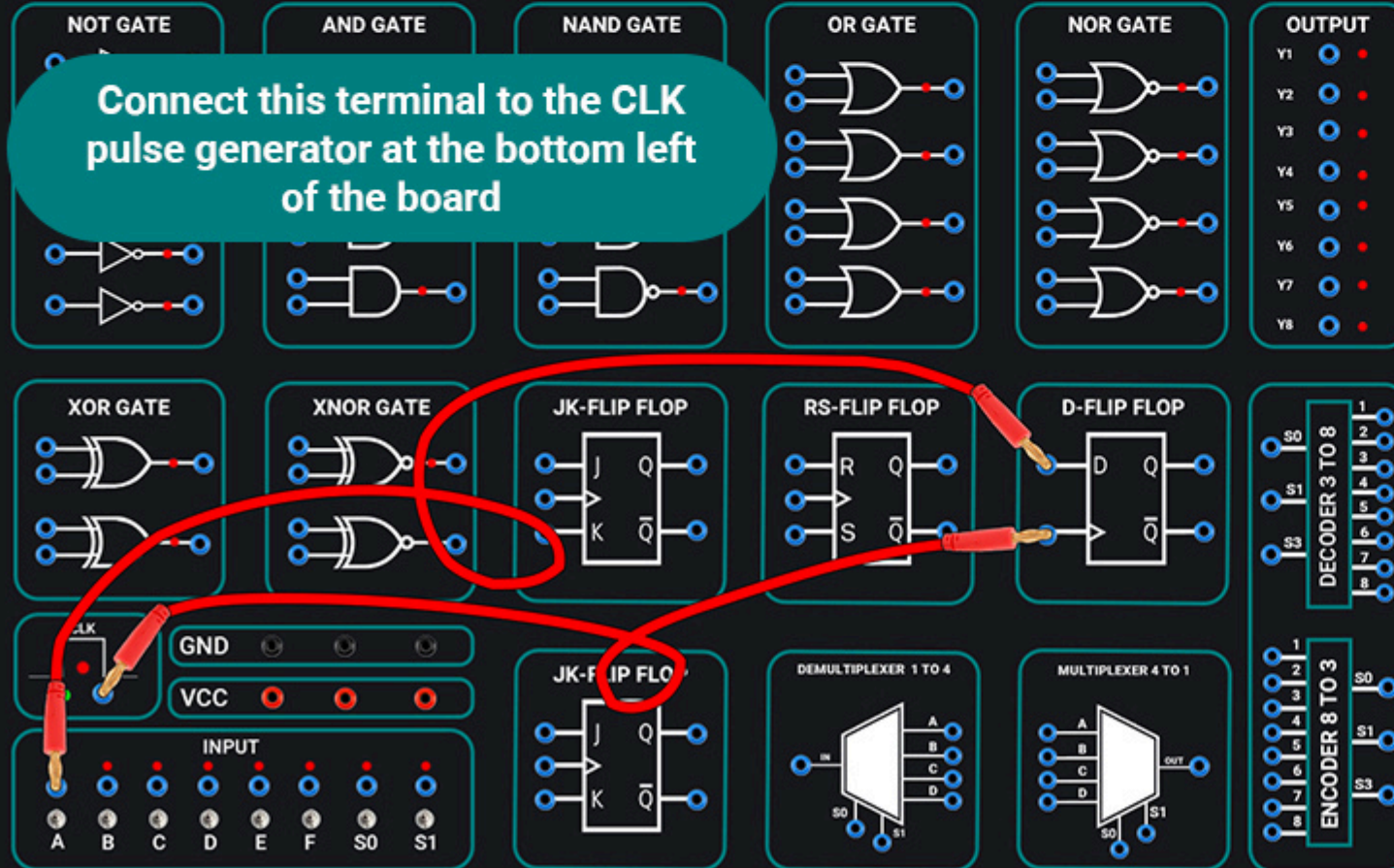
Connect the D input terminal (blue pin on the left) to switch A in the INPUT section



2

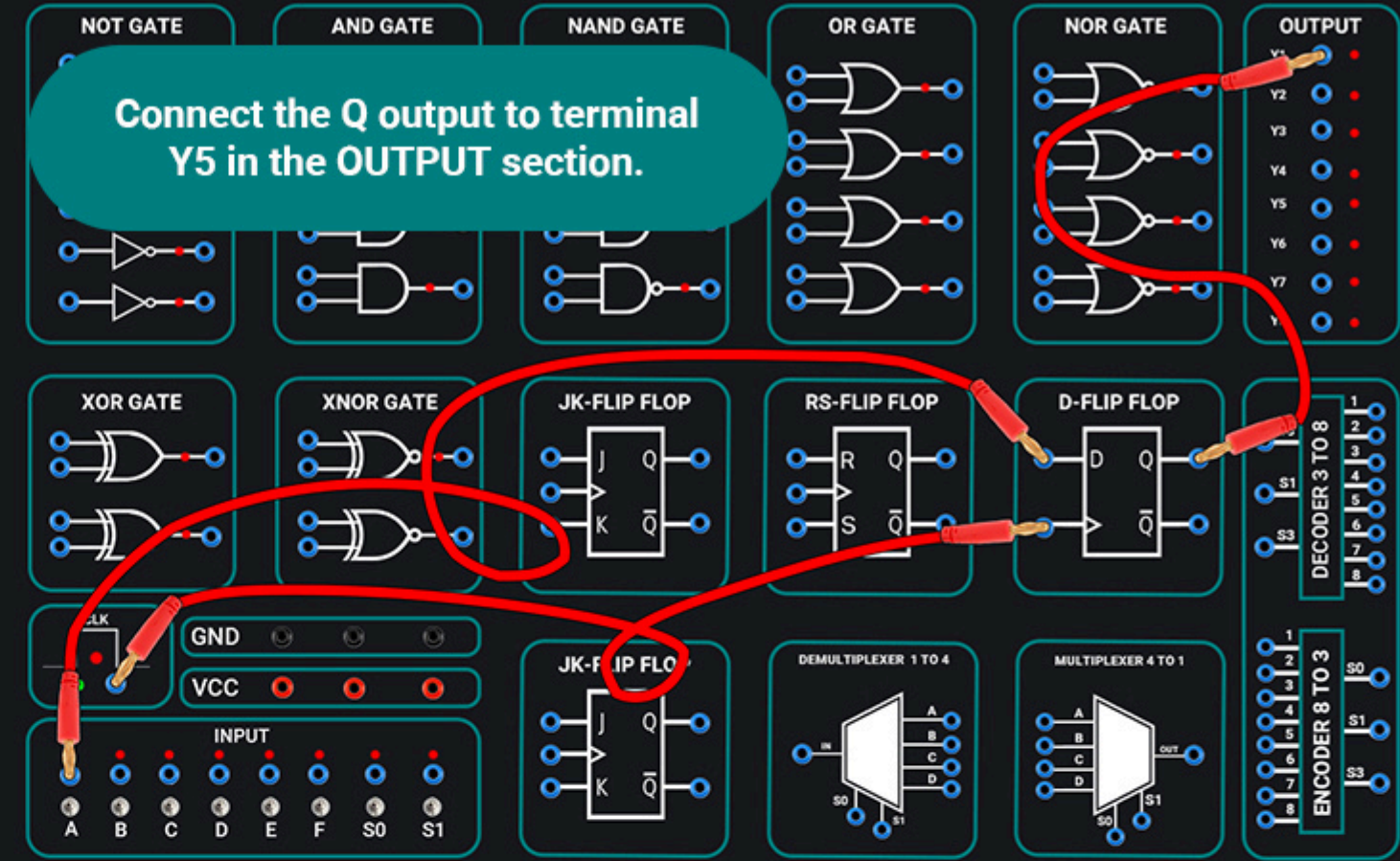
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Connect this terminal to the CLK pulse generator at the bottom left of the board



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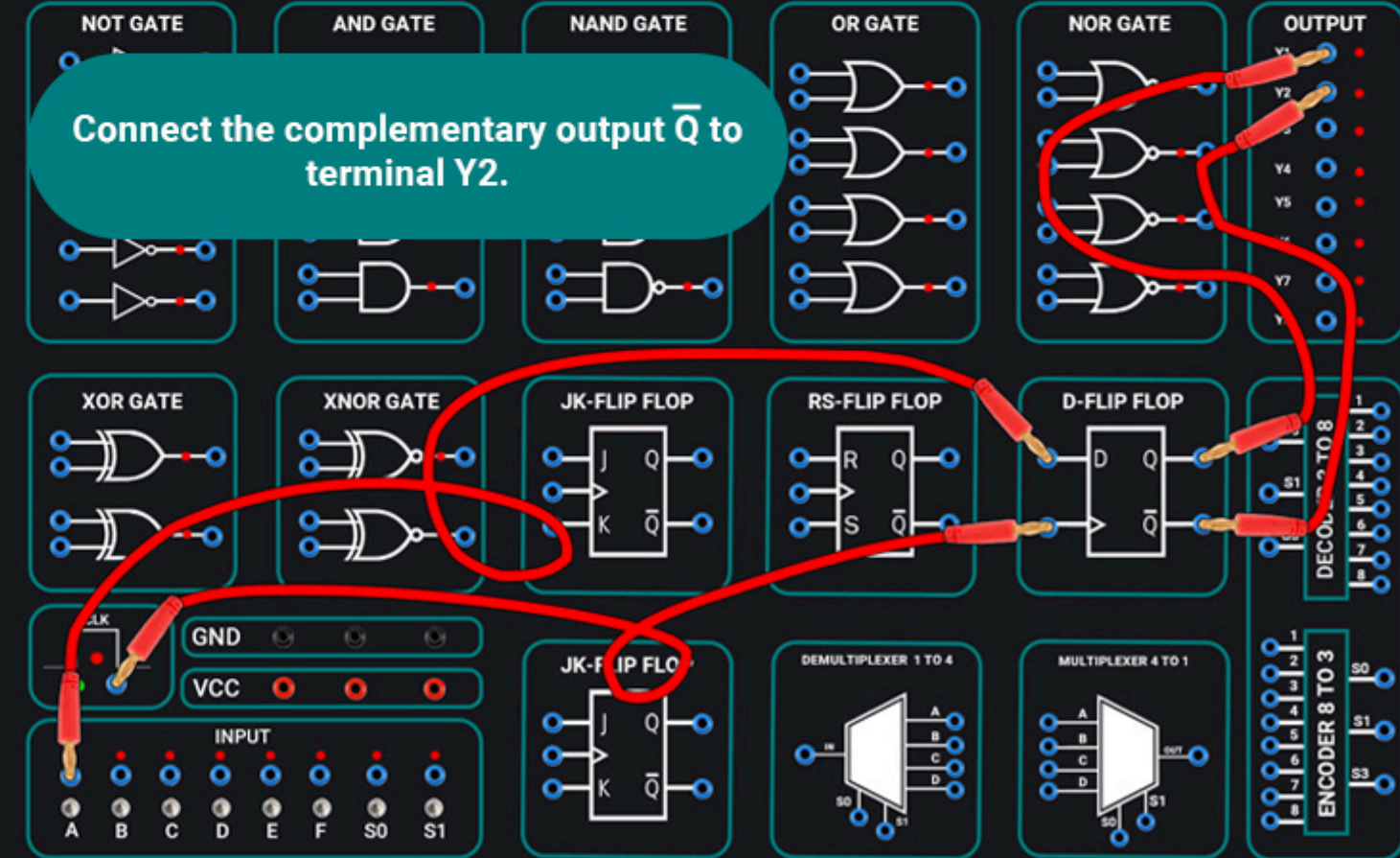
Connect the Q output to terminal Y5 in the OUTPUT section.



3

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Connect the complementary output $\bar{Q}$ to terminal Y2.



4

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